

EXAM 7 – SPRING 2015

9. (1.5 points)

Given the following portfolio characteristics and selected coefficients of variation (CoV):

<u>Portfolio</u>	<u>Portfolio Size (\$000,000)</u>	<u>Length of Claim Run Off (in years)</u>	<u>Selected Coefficients of Variation (CoV)</u>	
			<u>Outstanding Claim Liability</u>	<u>Premium Liability</u>
A	50	20	5.5%	y
B	300	20	x	7.0%
C	300	5	0.5%	0.3%

- Event risk is not material.

a. (0.75 point)

Select a value for x and explain why it is reasonable within the scope of internally benchmarking independent risk.

b. (0.75 point)

Select a value for y and explain why it is reasonable within the scope of internally benchmarking independent risk.

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QUESTION 9	
TOTAL POINT VALUE: 1.5	LEARNING OBJECTIVE: A5: Describe the various sources of risk and uncertainty that are associated with the determination of reserves.
SAMPLE ANSWERS	
Part a: 0.75 point	
<i>Sample Answer 1</i> Since Portfolio B has a very long claim run-off time, the Premium Liability COV should be higher than the OCL COV. Moreover, Portfolio B is larger (in size) than Portfolio A, which is also having the same length of claim runoff years. Thus the OCL COV for A is larger than the OCL COV for B. $5.5\% > x$, $7.0\% > x$ In addition, OCL COV for B is longer than OCL COV for C, since they are the same size, and C has a much shorter runoff time than B $\rightarrow x > 0.5\%$ Select $x = 5\%$ <i>Sample Answer 2</i> 5.5% - since the tail of claims matches A (C is a lot quicker, so lower CV), it would be an appropriate CV to account for the uncertainty.	
Part b: 0.75 point	
<i>Sample Answer 1</i> PL COV(A) > PL COV(B) (since A is smaller than B, but with the same runoff period) PL COV(A) > OCL COV(A) (more uncertainty for PL in long tail lines) $Y > 7\% \rightarrow$ select $y = 7.5\%$ <i>Sample Answer 2</i> A has smaller size & longer runoff length than C $\rightarrow$ that y should definitely be higher than 0.3%. Smaller book + same runoff length than B $\rightarrow y$ should be higher than 7% I choose y to be 10% because it is longer tailed & smaller sized <i>Sample Answer 3</i> $Y = 7.0\%$ because this matches portfolio B which has a similar claim runoff length. Premium liability is risk that premiums written will not cover losses, and these two appear to write similar length (likely liability) coverage	
EXAMINERS' REPORT	
- Candidates generally performed well with on this question, with a majority of candidates receiving full credit. Candidates were expected to understand the relationships between length of claim runoff and portfolio size and how that affects variability. - Although candidates were successful overall, candidates earned full credit on part a more frequently than on part b. Some candidates struggled with how premium liability COVs are affected by length of claim runoff and portfolio size more than they did with the outstanding claim liability COV. Some candidates explained how the mean of the OCL or the PL were impacted rather than the COV of the OCL or PL. Other candidates thought that	

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a lower portfolio size meant there was less premium liability variability as opposed to more variability.
Part a
• Candidates performed well on part a, with a majority of candidates receiving full credit. • The candidate was expected to understand how the COV for OCL is impacted by length of claim runoff and portfolio size. • In order to receive full credit, the candidate was expected to provide an acceptable value for x and explain why it is acceptable in relation to internal benchmarks, and why we would expect x to differ from the benchmarks. • The most common mistake was selecting a value for x greater than 5.5% – some candidates mistakenly thought a larger portfolio size increases the coefficient of variation – the mean of the OCL is expected to increase, but we would expect the variability as a percentage of the mean to actually go down as the volatility due to random effects decreases. • Some candidates explained how the COV should relate to the benchmarks but failed to actually provide a value. • Other candidates provided correct values but didn't explain why they were reasonable in the context of the internal benchmarks.
Part b
• Candidates performed well on part b, with a majority of candidates receiving full credit. • Candidates were expected to understand how the COV for PL is impacted by length of claim runoff and portfolio size. • In order to receive full credit, candidates were expected to provide an acceptable value for y and explain why it is acceptable in the context of internal benchmarks, and why we would expect y to differ from the benchmarks. • The most common mistake was selecting a value for y less than 7% – some candidates mistakenly thought a larger portfolio size increases the coefficient of variation – the mean of the PL is expected to increase, but we would expect the variability as a percentage of the mean to actually go down as the volatility due to random effects decreases. • Some candidates mistakenly thought that for long-tailed lines, $\text{COV}(\text{OCL}) > \text{COV}(\text{PL})$. They had this relationship reversed, we would actually expect $\text{COV}(\text{PL})$ to be greater than $\text{COV}(\text{OCL})$ for long-tailed lines. • Some candidates explained how the COV should relate to the benchmarks but failed to actually provide a value. • Other candidates provided reasonable values but did not explain why these were reasonable in the context of the internal benchmarks.